

Natural Language Processing

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Natural language processing (NLP) tasks

NLP is the process through which AI is taught to **understand the rules and syntax of language**, programmed to **develop complex algorithms to represent those rules**, and then made to **use those algorithms to carry out specific tasks** like these.



Language generation



Answering questions



Text classification



Sentiment analysis



Machine translation

► Why NLP?

- Humans speak different languages in many forms: text, speech, emojis!
- Computers don't "understand" — they deal with 1s and 0s.
- NLP bridges the gap between human language and machines.

► Examples:

- Google Translate
- ChatGPT
- Siri/Alexa
- Email spam filters

► Goal: Help computers understand, interpret, and even generate human language.

By the end of this session, you'll be able to:

- ▶ Understand key concepts and challenges in NLP.
- ▶ Know how RNN, LSTM, GRU work (without getting stuck in math!).
- ▶ Explore how attention and transformers changed the NLP game.
- ▶ Get hands-on intuition to build simple NLP models.
- ▶ Be ready to dive into modern language models (like GPT, BERT, etc.)

NLP: Basics

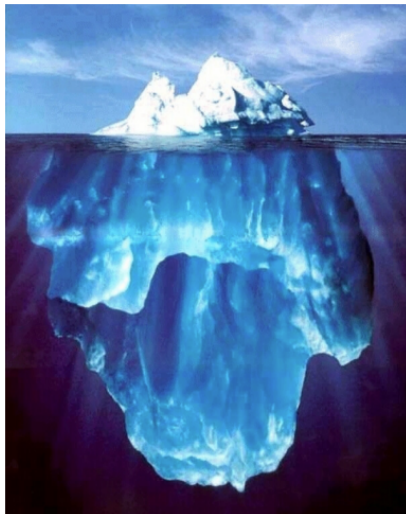
What is NLP?

- ▶ NLP stands for Natural Language Processing.
- ▶ It helps computers understand and work with human languages, like English or Arabic.
- ▶ With NLP, computers can read, listen, write, or even talk like humans.
- ▶ Examples: chatbots, language translators, voice assistants.

An iceberg is a large piece of freshwater ice that has broken off from a snow-formed glacier or ice shelf and is floating in open water.



.. But Language is Complex



An iceberg is a large piece of freshwater ice that has broken off from a snow-formed glacier or ice shelf and is floating in open water.

Teaching machines to **understand and generate human language**

- ▶ **Text preprocessing:**

Cleaning and preparing raw text data (removing noise, tokenization, normalization).

- ▶ **Understanding meaning:**

Extracting meaning from text using techniques like part-of-speech tagging, named entity recognition, and sentiment analysis.

- ▶ **Language modeling:**

Building models that can predict or generate text, such as autocomplete or next-word prediction.

- ▶ **Translation, summarization, etc.:**

Enabling applications like machine translation, text summarization, question answering, and more.

Text data is everywhere: tweets, reviews, articles, chats! NLP helps us make sense of this vast information.

Task	What It Does	Example
Tokenization	Split text into words	"I love NLP" → ["I", "love", "NLP"]
POS Tagging	Label grammar tags	"Dogs bark" → [Noun, Verb]
Named Entity Recognition	Find names, places, etc.	"Christopher Nolan lives in Los Angeles"
Sentiment Analysis	Detect mood	"This movie was amazing!" → Positive
Machine Translation	Language to language	English → French

Core NLP Tasks and Examples

► **Lowercase everything:**

Example: "NLP" → "nlp"

► **Removing stop words:**

Eliminate common words that carry little meaning (e.g., "is", "the", "and") to focus on important content.

► **Stemming/Lemmatization:**

Reduce words to their root or base form.

Example: "running" → "run"

► **Vectorization:**

Transform words or documents into numerical representations for machine learning models:

- **Bag of Words:** Counts word occurrences in a document.
- **TF-IDF (Term Frequency-Inverse Document Frequency):** Weighs words by importance across documents.
- **Word2Vec:** Learns dense vector representations capturing word meaning and context.

What is Vocabulary in NLP?

- ▶ **Vocabulary** = All unique words in your dataset
- ▶ Example: "I love NLP and NLP loves me" →
Vocabulary = {"I", "love", "NLP", "and", "loves", "me"}
- ▶ More data = Bigger vocabulary = Harder to process!

Tip: Rare words may not help; common words may not mean much.

Traditional approach: One-hot encoding

- ▶ Example: "NLP" $\rightarrow [0, 0, 1, 0, 0, 0, 0, 0 \dots]$

Problem:

- ▶ High-dimensional (thousands of words!)
- ▶ Sparse (mostly 0s)
- ▶ No meaning in structure (no relation between "king" and "queen")
- ▶ Not efficient for learning

Feature extraction = Turning text into numbers

- ▶ Count how often each word appears (**Term Frequency**)

Example:

Text	"great product"	"bad product"
Word: "great"	1	0
Word: "bad"	0	1

Use this to find patterns in sentiment, spam, etc.

Suppose you're classifying reviews:

Positive Reviews: ["amazing", "good", "great"]

Negative Reviews: ["bad", "awful", "terrible"]

Count how often each word appears in each class.

Example table:

Word	Positive Count	Negative Count
good	20	1
bad	1	30

Helps models detect the “tone” (**sentiment clues**) of new text.

- ▶ **Bag of Words (BoW)**: Just counts word frequencies in each document.
- ▶ **TF-IDF (Term Frequency-Inverse Document Frequency)**: Adjusts for how “unique” or important a word is in a document compared to all documents.
 - Words like "the", "is" are less important.
- ▶ **Word Embeddings (later)**: Add meaning and capture relationships between words (e.g., similarity, analogy).

BoW ignores word order: "not good" = "good"

N-grams fix this:

- ▶ **1-gram:** "I", "love", "cats"
- ▶ **2-gram:** "I love", "love cats"

Helps detect phrases:

- ▶ "not good" $\neq$ "very good"

Bi-grams and tri-grams make models more context-aware!

NLP is all about uncertainty!

We ask: **What is the probability that this sentence is positive?**

Example:

- ▶ $P(\text{Positive} \mid \text{"great product"}) = 0.92$
- ▶ $P(\text{Negative} \mid \text{"great product"}) = 0.08$

Bayes' Rule:

$$P(\text{Class} \mid \text{Words}) = \frac{P(\text{Words} \mid \text{Class}) \cdot P(\text{Class})}{P(\text{Words})}$$

- ▶ Predict class (e.g., spam or not) given the words
- ▶ Simple, powerful tool used in Naive Bayes classifiers
- ▶ Helps models reason with uncertainty

Simple, fast probabilistic model

Assumes:

- ▶ Words are independent given the class (a “naive” assumption)

Example:

- ▶ "This movie is awesome" $\rightarrow P(\text{Positive})$ high
- ▶ "This movie is boring" $\rightarrow P(\text{Negative})$ high

Works well with BoW, TF-IDF, and N-grams

We're already familiar with probability-based classifiers, and have seen logistic regression applied in computer vision tasks.

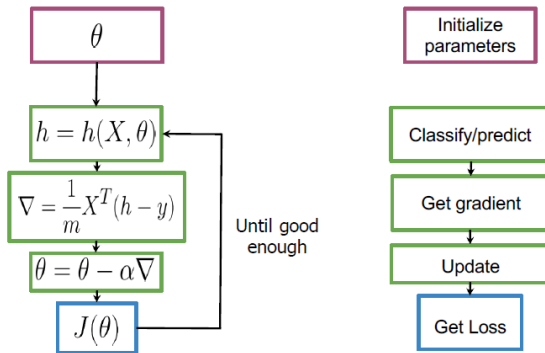
- ▶ Here, logistic regression takes word features (from BoW or TF-IDF)
- ▶ It learns weights to predict if text belongs to a certain class, such as:
 - Positive or Negative
 - Spam or Not Spam
- ▶ Outputs a probability:

$$P(\text{Class} = 1 \mid \text{features}) = \sigma(w_1x_1 + w_2x_2 + \dots + b)$$

- ▶ Words like “awesome” get high weights for positive class

Intuition:

- ▶ Each word contributes a “vote” toward a class.
- ▶ The model learns which words are most important for classification.



Training a Logistic Regression Model

Unlike BoW, word vectors (embeddings) capture meaning

Words are mapped to dense vectors in low-dimensional space.

Example: 300-dimensional vector: "cat" $\rightarrow$ [0.1, -0.3, ..., 0.2]

Similar words have similar vectors:

- ▶ "king" and "queen" are close in vector space
- ▶ Captures relationships and analogies (e.g., "man" : "woman" :: "king" : "queen")

Word embeddings add context and meaning to NLP models!

- ▶ **Word2Vec (Google):**
Trains using context words (“You shall know a word by the company it keeps”)
- ▶ **GloVe (Stanford):**
Uses word co-occurrence in large corpus
- ▶ **FastText (Facebook):**
Captures subword info, works better for rare words

Used for initializing deep models like RNNs, Transformers

- ▶ **Vocabulary and word counts** are key features for representing text.
- ▶ **Sparse data** (like one-hot vectors) limits early NLP models.
- ▶ **Naive Bayes** and **Logistic Regression** are fast and effective for text classification.
- ▶ **N-grams** add word order and context.
- ▶ **Word vectors** (embeddings) bring semantic meaning to models.

Now, let's dive into neural networks for language...

NLP: Recurrent Neural Networks (RNNs)

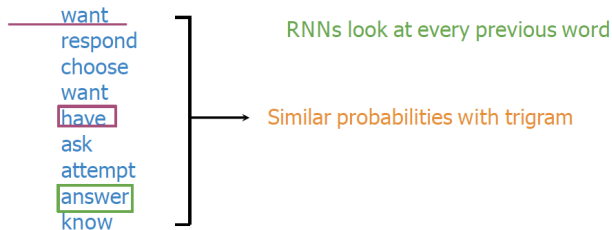
► Bag-of-Words and n-gram models:

- Ignore word order and long-range dependencies.
- Fixed-size context window limits understanding of sequence.
- Consume more memory and computational resources for larger contexts.

► Feedforward Neural Networks:

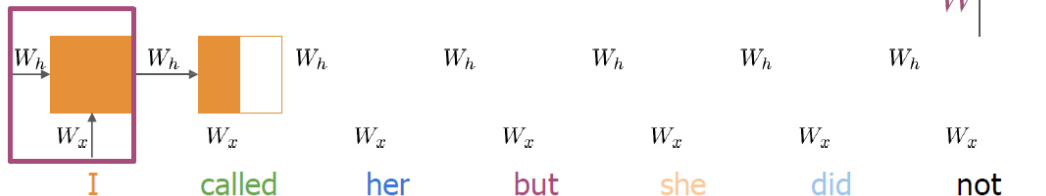
- Cannot handle variable-length input sequences.
- Lack memory of previous inputs.

Nour was supposed to study with me. I called her but she did not answer



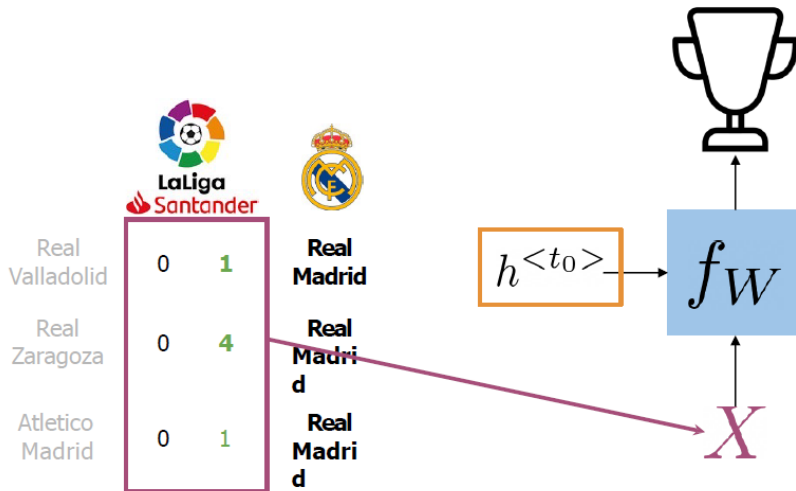
- ▶ Designed to process sequential data of arbitrary length.
- ▶ Maintain a hidden state to capture context and dependencies over time.
- ▶ Better suited for tasks like language modeling, translation, and sequence prediction.

I called her but she did not _____



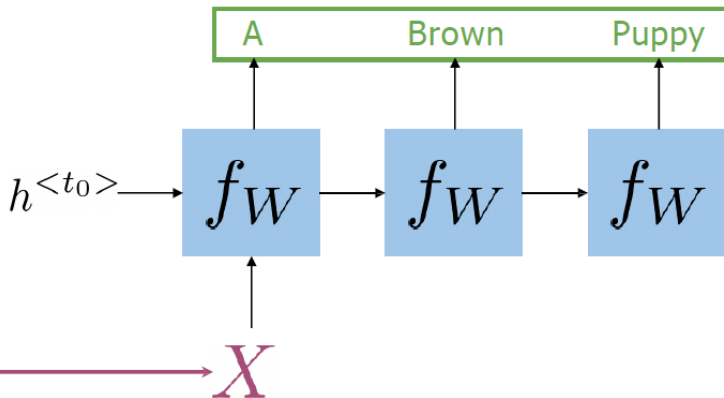
Learnable parameters

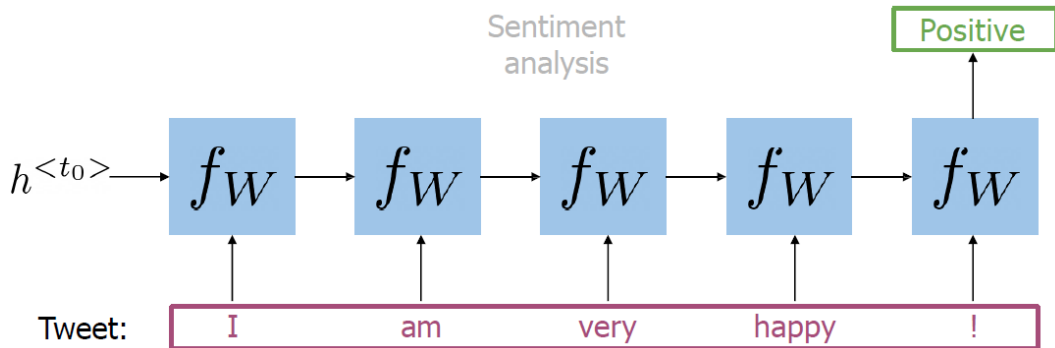
- ▶ RNNs model relationships among distant words in a sequence.
- ▶ Many computations in RNNs share parameters across time steps, enabling efficient learning and generalization.

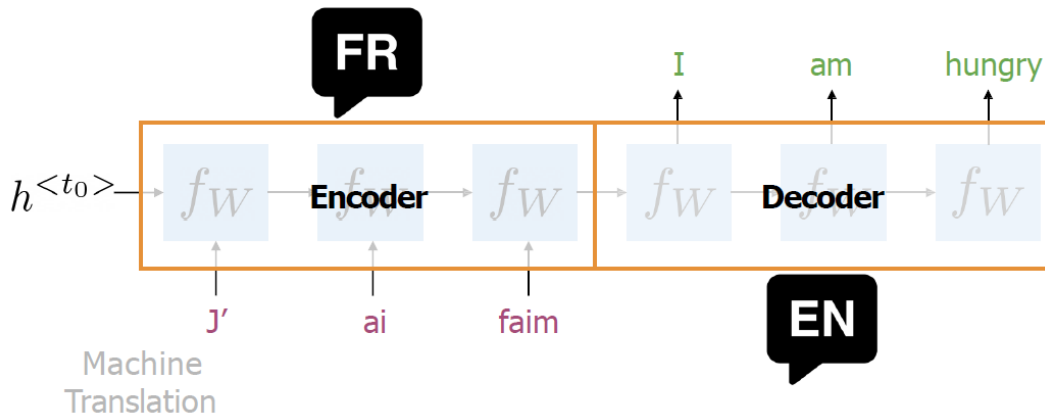


One to Many

Caption
generation







► Propagation through time:

$$h_t = \sigma(W_{xh}x_t + W_{hh}h_{t-1} + b_h)$$

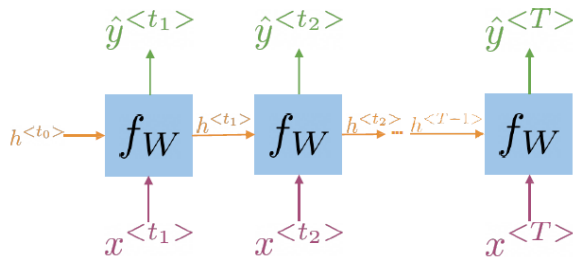
where x_t is the input at time t , h_{t-1} is the previous hidden state, W_{xh} and W_{hh} are weight matrices, b_h is the bias, and σ is an activation function (e.g., tanh or ReLU).

► Making predictions:

$$y_t = \phi(W_{hy}h_t + b_y)$$

where W_{hy} is the output weight matrix, b_y is the output bias, and ϕ is an activation function (e.g., softmax for classification).

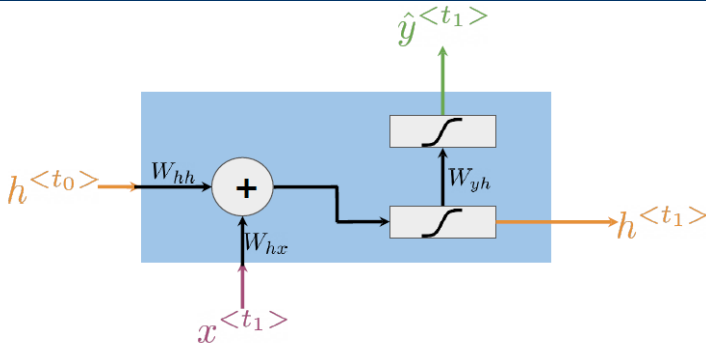
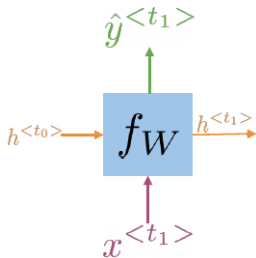
Key idea: The hidden state h_t acts as memory, allowing information to flow from previous time steps to influence current predictions.



$$h^{<t>} = g(W_h[h^{<t-1>}, x^{<t>}] + b_h)$$
$$\hat{y}^{<t>} = g(W_{yh}h^{<t>} + b_y)$$

$$h^{<t>} = g(W_{hh}h^{<t-1>} + W_{hx}x^{<t>} + b_h)$$

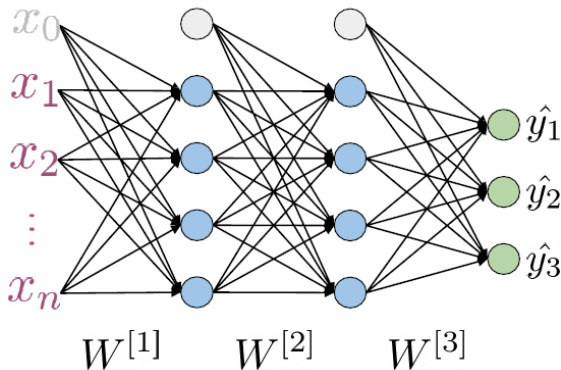
A Vanilla RNN (cont.)



$$h^{<t>} = g(W_{hh}h^{<t-1>} + W_{hx}x^{<t>} + b_h)$$
$$\hat{y}^{<t>} = g(W_{yh}h^{<t>} + b_y)$$

- Hidden states propagate information through time.

- ▶ Basic recurrent units have two inputs at each time: $x^{(t)}$ (current input) and $h^{(t-1)}$ (previous hidden state).

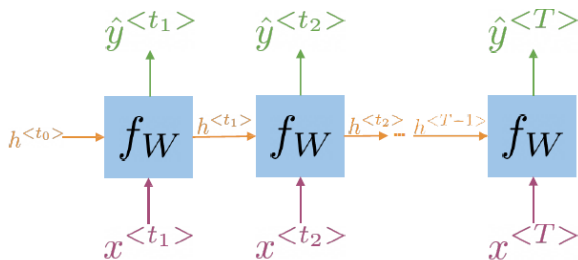


K - classes or possibilities

$$J = - \sum_{j=1}^K y_j \log \hat{y}_j$$

Either 0 or 1

Looking at a single example (x, y)



$$h^{<t>} = g(W_h[h^{<t-1>}, x^{<t>}] + b_h)$$

$$\hat{y}^{<t>} = g(W_{y_h}h^{<t>} + b_y)$$

$$J = -\frac{1}{T} \sum_{t=1}^T \sum_{j=1}^K y_j^{<t>} \log \hat{y}_j^{<t>}$$

Average with respect to time

For RNNs the loss function is just an average through time!



NLP: **Gated Recurrent Unit (GRU)**

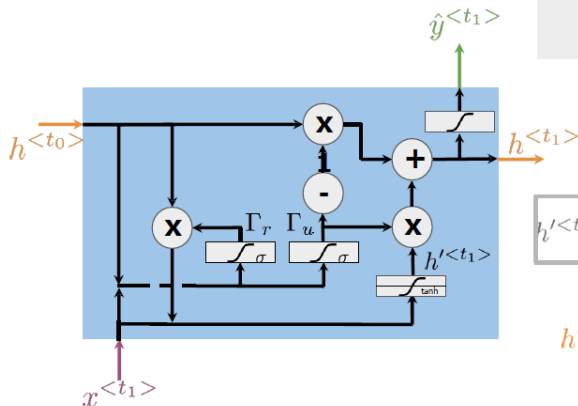
- ▶ GRU is a type of RNN designed to handle long-range dependencies.
- ▶ Combines the forget and input gates into a single update gate.
- ▶ Uses a reset gate to control how much past information to forget.

"**Ants** are really interesting. They are everywhere."

↓
Plural



Relevance and update gates to remember important prior information



Gates to keep/update relevant information in the hidden state

$$\begin{aligned}\Gamma_r &= \sigma(W_r[h^{<t_0>}, x^{<t_1>}] + b_r) \\ \Gamma_u &= \sigma(W_u[h^{<t_0>}, x^{<t_1>}] + b_u)\end{aligned}$$

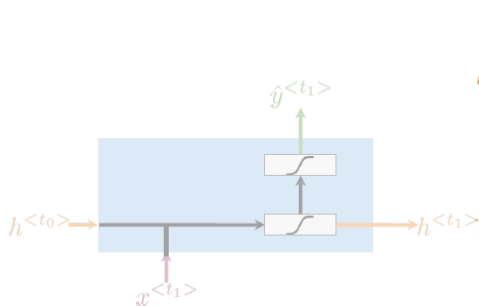
$$h'^{<t_1>} = \tanh(W_h[\Gamma_r * h^{<t_0>}, x^{<t_1>}] + b_h)$$

Hidden state candidate

$$h^{<t_1>} = (1 - \Gamma_u) * h^{<t_0>} + \Gamma_u * h'^{<t_1>}$$

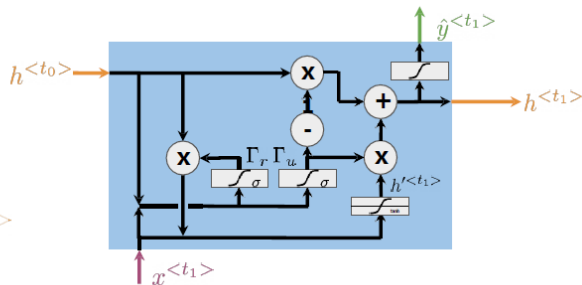
$$\hat{y}^{<t_1>} = g(W_y h^{<t_1>} + b_y)$$

Vanilla RNN vsGRUs



$$h^{<t>} = g(W_h[h^{<t-1>}, x^{<t>}] + b_h)$$

$$\hat{y}^{<t>} = g(W_y h^{<t>} + b_y)$$



$$\Gamma_u = \sigma(W_u[h^{<t_0>}, x^{<t_1>}] + b_u)$$

$$\Gamma_r = \sigma(W_r[h^{<t_0>}, x^{<t_1>}] + b_r)$$

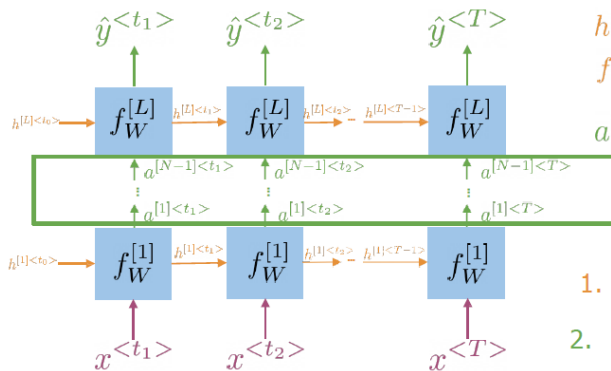
$$h'^{<t_1>} = \tanh(W_h[\Gamma_r * h^{<t_0>}, x^{<t_1>}] + b_h)$$

$$h^{<t_1>} = (1 - \Gamma_u) * h^{<t_0>} + \Gamma_u * h'^{<t_1>}$$

$$\hat{y}^{<t_1>} = g(W_y h^{<t_1>} + b_y)$$

- ▶ GRUs “decide” how to update the hidden state:
 - The update gate controls the degree to which the hidden state is updated with new information.
 - The reset gate determines how much of the previous hidden state to forget.
 - This gating mechanism allows the GRU to adaptively capture dependencies of varying lengths.

- ▶ GRUs help preserve important information:
 - By selectively updating and resetting, GRUs can retain relevant information over long sequences.
 - This helps mitigate the vanishing gradient problem common in standard RNNs.
 - As a result, GRUs are effective for tasks requiring memory of long-term context.



$$h^{[l]<t>} = f^{[l]}(W_h^{[l]}[h^{[l]<t-1>}, a^{[l-1]<t>}] + b_h^{[l]})$$

$$a^{[l]<t>} = f^{[l]}(W_a^{[l]}h^{[l]<t>} + b_a^{[l]})$$

Intermediate
layers and
activations

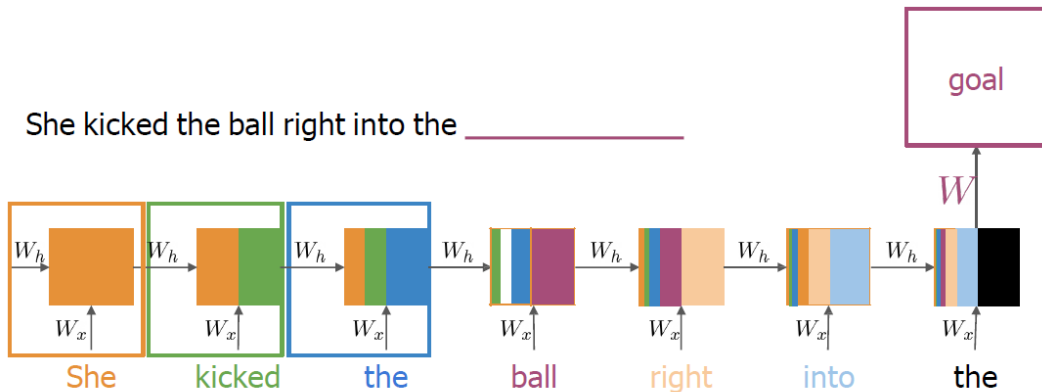
1. Get hidden states for current layer
2. Pass the activations to the next layer

Deep RNNs have more than one layer, which helps in complex tasks.

- ▶ Captures dependencies within a short range.
- ▶ Takes up less RAM than other n-gram models.

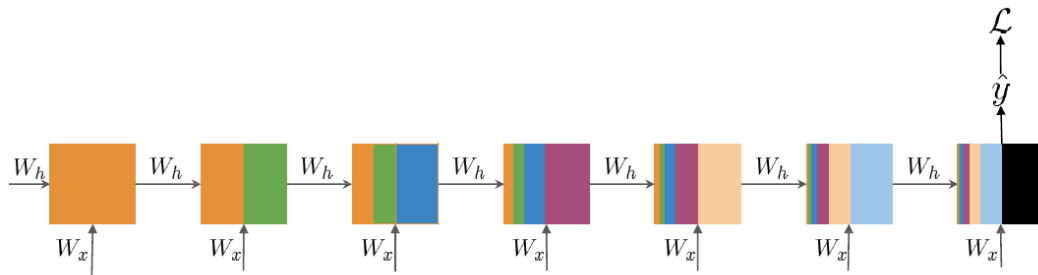
- ▶ Struggles to capture long-term dependencies.
- ▶ Prone to vanishing or exploding gradients.

She kicked the ball right into the _____



Learnable parameters

Backpropagation Through Time (BPTT)



W_x
 W_h $\rightarrow$ Same at every step

$$\frac{\partial L}{\partial W_h} \propto \sum_{1 \leq k \leq t} \left(\prod_{t \geq i > k} \frac{\partial h_i}{\partial h_{i-1}} \right) \frac{\partial h_k}{\partial W_h}$$

Gradient is proportional to a sum of partial derivative products

$$\frac{\partial L}{\partial W_h} \propto \sum_{1 \leq k \leq t} \left(\prod_{t \geq i > k} \frac{\partial h_i}{\partial h_{i-1}} \right) \frac{\partial h_k}{\partial W_h} \rightarrow \text{Contribution of hidden state } k$$

Length of the product proportional to
how far k is from t

$$\frac{\partial h_t}{\partial h_{t-1}} \frac{\partial h_{t-1}}{\partial h_{t-2}} \frac{\partial h_{t-2}}{\partial h_{t-3}} \frac{\partial h_{t-3}}{\partial h_{t-4}} \frac{\partial h_{t-4}}{\partial h_{t-5}} \frac{\partial h_{t-5}}{\partial h_{t-6}} \frac{\partial h_{t-6}}{\partial h_{t-7}} \frac{\partial h_{t-7}}{\partial h_{t-8}} \frac{\partial h_{t-8}}{\partial h_{t-9}} \frac{\partial h_{t-9}}{\partial h_{t-10}} \frac{\partial h_{t-10}}{\partial W_h}$$

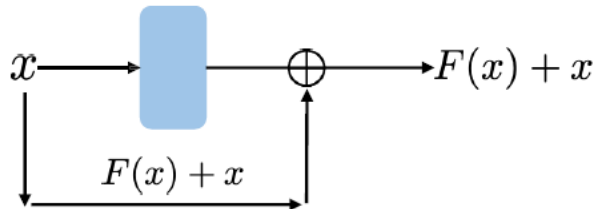
Contribution of hidden state $t-10$

- Identity RNN with ReLU activation

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Gradient clipping

- Skip connections



NLP: Long Short-Term Memory (LSTM)

- ▶ LSTM is a type of RNN designed to overcome the vanishing gradient problem.
- ▶ It uses a memory cell to maintain information over long periods.
- ▶ Learns when to remember and when to forget.
- ▶ **Basic anatomy:**
 - A cell state
 - A hidden state
 - Multiple gates
- ▶ LSTM has three gates: input gate, forget gate, and output gate.
- ▶ Gates allow gradients to avoid vanishing and exploding.

Starting point with some irrelevant information



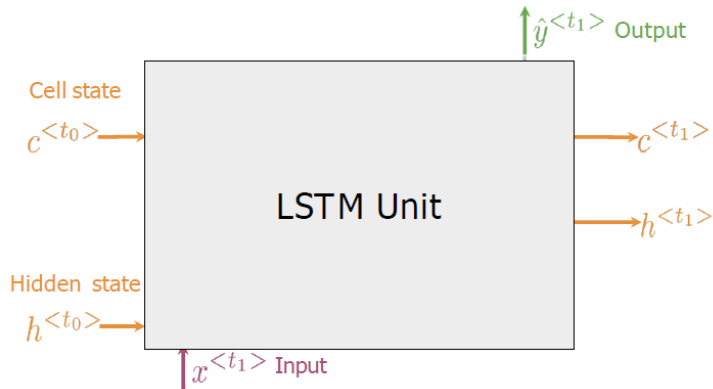
Cell and Hidden States

Discard anything irrelevant

Add important new information

Produce output

→ Gates



1. Forget Gate:

information that is no longer important

2. **Input Gate:** information to be stored

3. Output Gate:

information to use at current step

Next-character
prediction



Chatbots



Music
composition



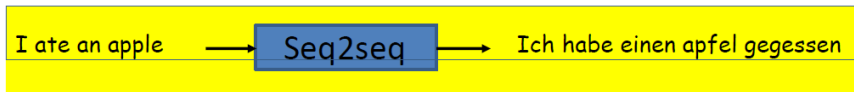
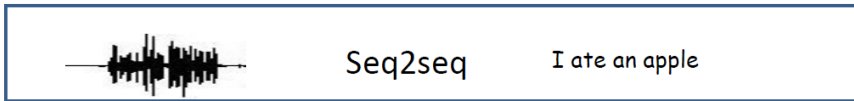
Image
captioning



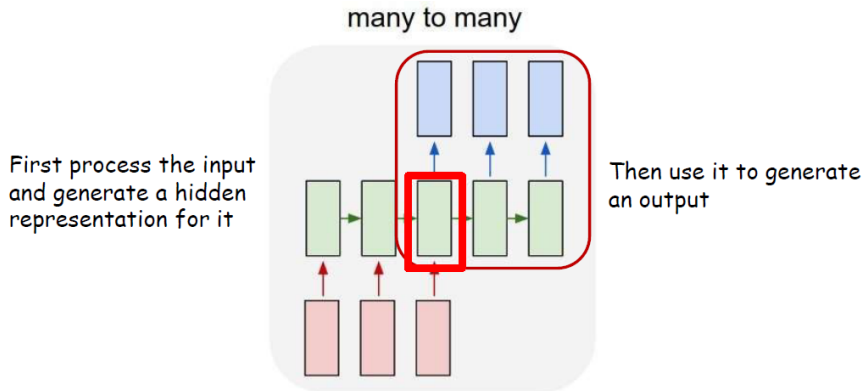
Speech
recognition



NLP: **Attention Mechanism**

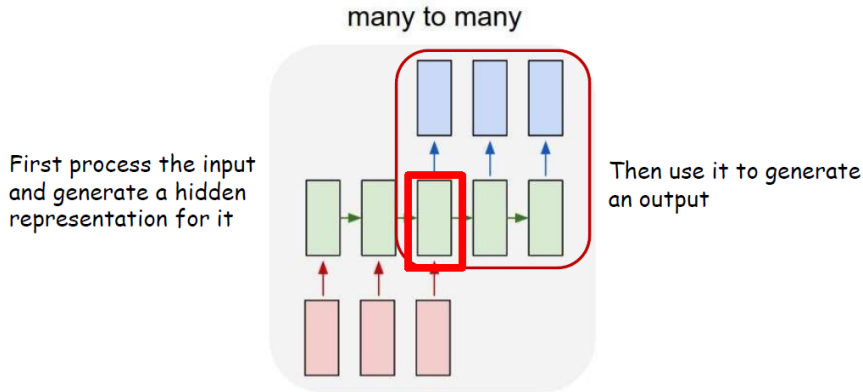


- Sequence goes in, sequence comes out
- No notion of “time synchrony” between input and output
 - May even not even maintain order of symbols
 - E.g. “I ate an apple” → “Ich habe einen apfel gegessen”
- Or even seem related to the input
 - E.g. “My screen is blank” → “Please check if your computer is plugged in.”



Delayed Sequence to Sequence Problem:

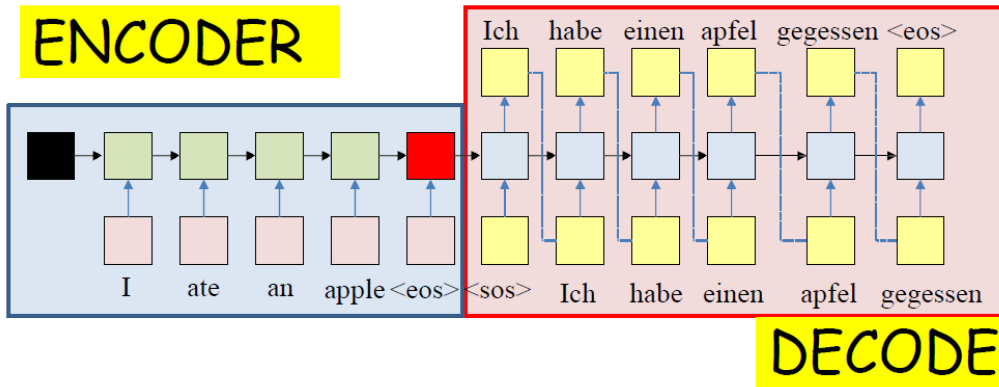
Problem: Each word that is output depends only on the current hidden state, and not on previous outputs.



Delayed Sequence to Sequence Problem:

Delayed *self-referencing* sequence-to-sequence: Each word that is output depends on the current hidden state, and also on previous outputs.

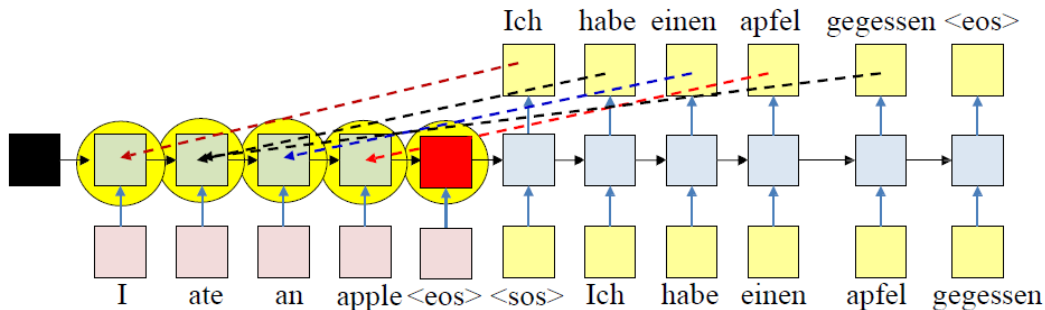
The “simple” translation model



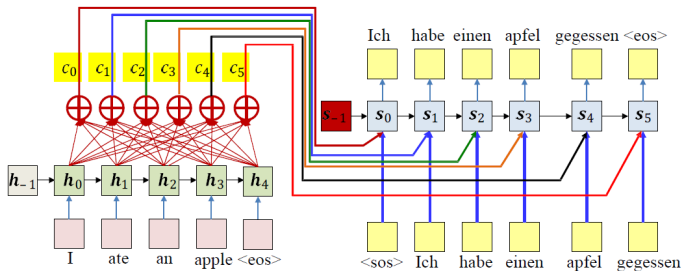
The recurrent structure that:

- ▶ extracts the hidden representation from the input sequence is the **encoder**.
- ▶ utilizes this representation to produce the output sequence is the **decoder**.

Problems with the “simple” translation model

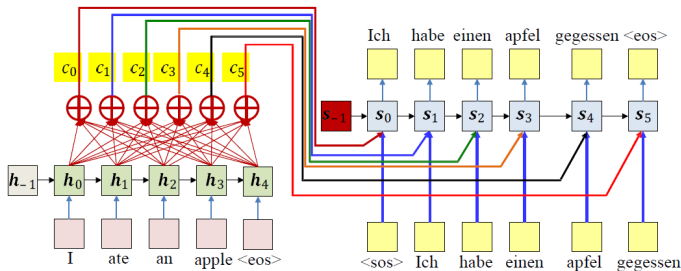


- ▶ The model has to compress all the information from the input sequence into a single fixed-size vector.
- ▶ This can lead to loss of important information, especially for long sequences.
- ▶ The model struggles with long-range dependencies and context retention.



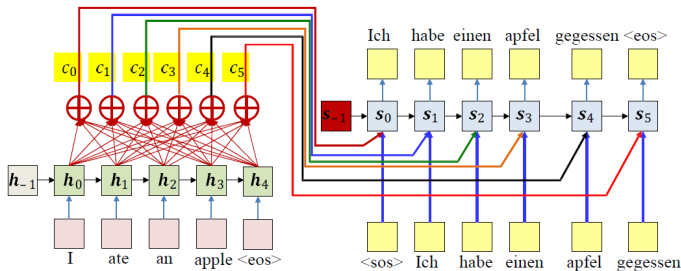
$$c_t = \frac{1}{N} \sum_{i=1}^N w_i(t) h_i$$

- **Attention weights:** The weights $w_i(t)$ are dynamically computed as functions of the decoder state and the encoder outputs.



$$c_t = \frac{1}{N} \sum_i^N w_i(t) h_i$$

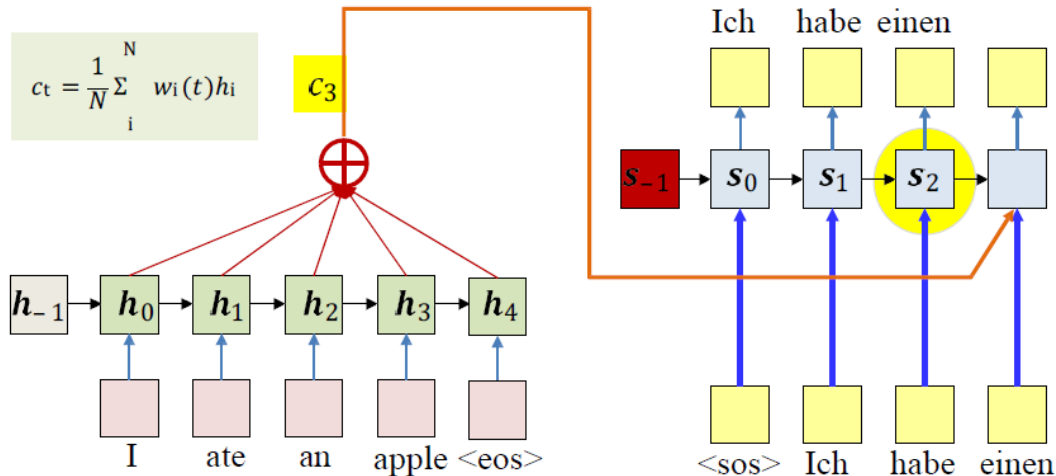
- **Attention weights:** The weights $w_i(t)$ are dynamically computed as functions of the decoder state and the encoder outputs.
- **Intuition:** If the model is well-trained, these weights will automatically “highlight” the most relevant parts of the input sequence for each output step.



$$c_t = \frac{1}{N} \sum_i^N w_i(t) h_i$$

- **Attention weights:** The weights $w_i(t)$ are dynamically computed as functions of the decoder state and the encoder outputs.
- **Intuition:** If the model is well-trained, these weights will automatically “highlight” the most relevant parts of the input sequence for each output step.
- **How are they computed?**

Attention Weights at time t



- **What do attention weights do?** They tell the model which input words to pay more attention to when making each output word.

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- ▶ The attention score is typically computed as:

$$w_i(t) = a(h_i, s_{t-1})$$

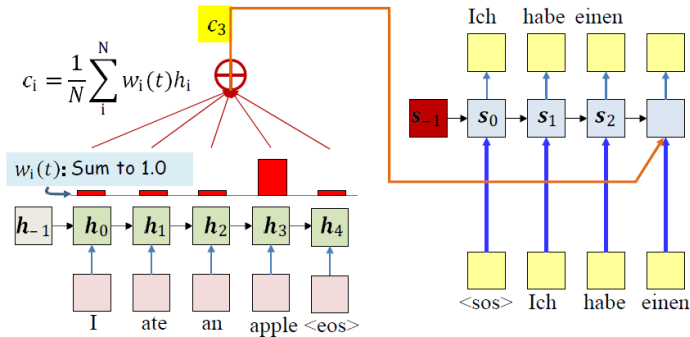
where $a(\cdot)$ is a function (such as a dot product, additive, or other scoring function).

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where $a(\cdot)$ is a function (such as a dot product, additive, or other scoring function).

- ▶ Sometimes, the input word at time t is also used, but often omitted for simplicity.



- The attention weights $w_i(t)$ must satisfy the following properties:
- They are non-negative: $w_i(t) \geq 0$ for all i .
 - They sum to 1: $\sum_{i=1}^n w_i(t) = 1$, where n is the length of the input sequence.

- ▶ These properties ensure that the attention weights can be interpreted as probabilities, allowing the model to focus on different parts of the input sequence.
- ▶ The non-negativity ensures that the model does not "ignore" any part of the input sequence, while the summation to 1 ensures that the model can distribute its attention across the input sequence.

Two-step computation of attention weights

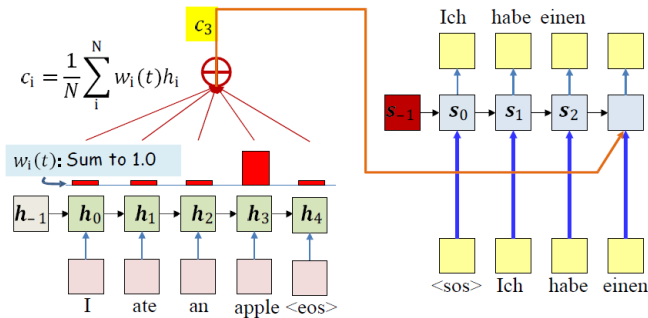
1. **Compute raw attention scores:** For each input position i , compute a score $e_i(t)$ (which can be positive or negative) using a scoring function:

$$e_i(t) = a(h_i, s_{t-1})$$

2. **Normalize with softmax:** Convert the raw scores into a probability distribution using the softmax function:

$$w_i(t) = \frac{\exp(e_i(t))}{\sum_{j=1}^n \exp(e_j(t))}$$

This ensures all $w_i(t) \geq 0$ and $\sum_{i=1}^n w_i(t) = 1$.



- Typical options for $g()$ (**variables in red must be learned**)

$$g(h_i, s_{t-1}) = h_i^T s_{t-1}$$

$$g(h_i, s_{t-1}) = h_i^T W_g s_{t-1}$$

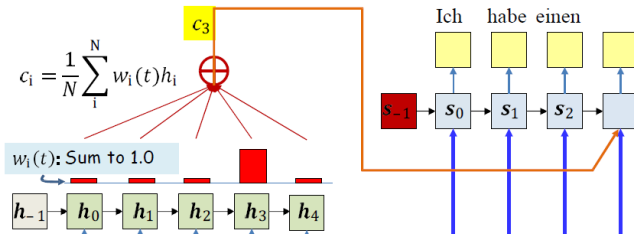
$$g(h_i, s_{t-1}) = v_g^T \tanh \left(W_g \begin{bmatrix} h_i \\ s_{t-1} \end{bmatrix} \right)$$

$$g(h_i, s_{t-1}) = \text{MLP}([h_i, s_{t-1}])$$

$$e_i(t) = g(h_i, s_{t-1})$$

$$w_i(t) = \frac{\exp(e_i(t))}{\sum_i \exp(e_i(t))}$$

Attention Weights at time t (cont.)



Let's consider a typical conversion process assuming this model as an example

- Typical options for $g()$ (**variables in red must be learned**)

$$g(h_i, s_{t-1}) = h_i^T s_{t-1}$$

$$g(h_i, s_{t-1}) = h_i^T \mathbf{w}_g s_{t-1}$$

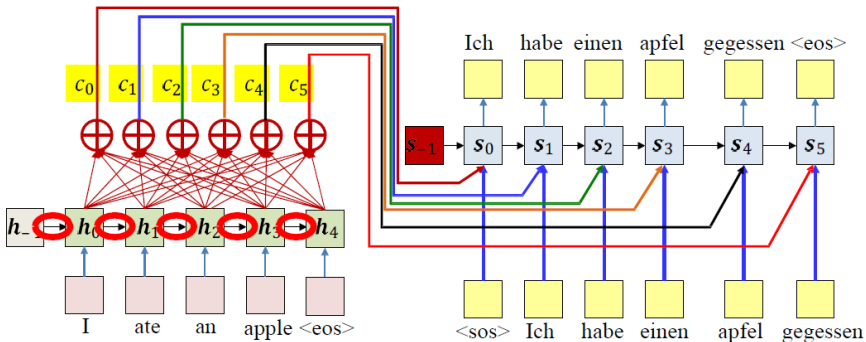
$$g(h_i, s_{t-1}) = \mathbf{v}_g^T \tanh\left(\mathbf{w}_g \begin{bmatrix} h_i \\ s_{t-1} \end{bmatrix}\right)$$

$$g(h_i, s_{t-1}) = \text{MLP}([h_i, s_{t-1}])$$

$$e_i(t) = g(h_i, s_{t-1})$$

$$w_i(t) = \frac{\exp(e_i(t))}{\sum_j \exp(e_j(t))}$$

Summary: Attention Models



- **Problem:** The decoder's hidden representation for each output word is influenced by *all* preceding words due to recurrence.
- The decoder effectively pays attention to the entire input sequence, not just a single word.
- If the decoder can automatically determine which input words to focus on at each step, recurrence in the input may not be strictly necessary.

NLP: **Transformers**

Attention Is All You Need

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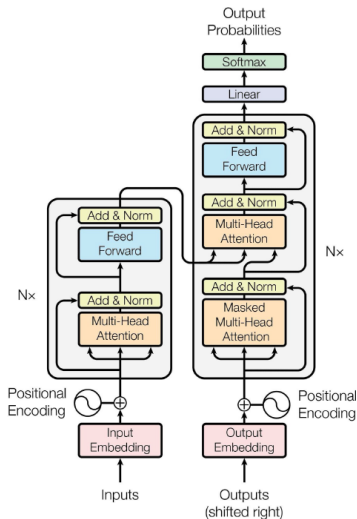
Aidan N. Gomez* †
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illia.polosukhin@gmail.com

Key Features:

- ▶ Transformers revolutionized NLP by using self-attention mechanisms.
- ▶ They process entire sequences simultaneously, unlike RNNs.
- ▶ Key components:
 - Multi-head self-attention
 - Positional encoding
 - Feed-forward neural networks
- ▶ Transformers can handle long-range dependencies effectively.



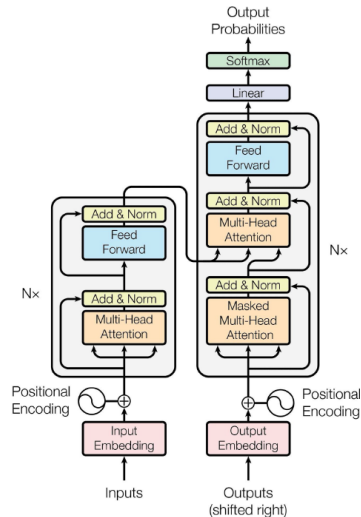
Targets

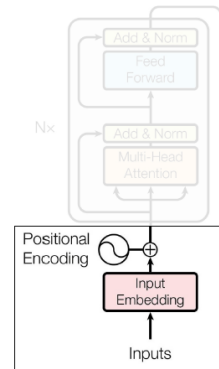
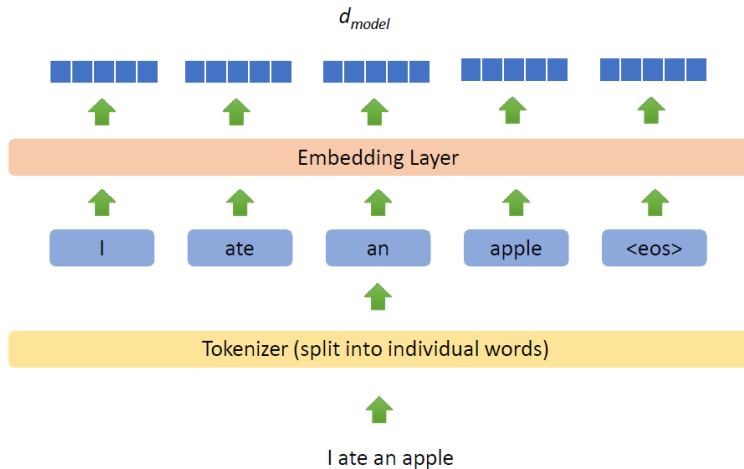
Ich habe einen Apfel
gegessen



Inputs

I ate an apple

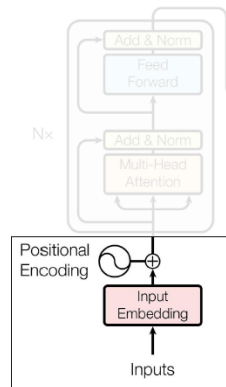
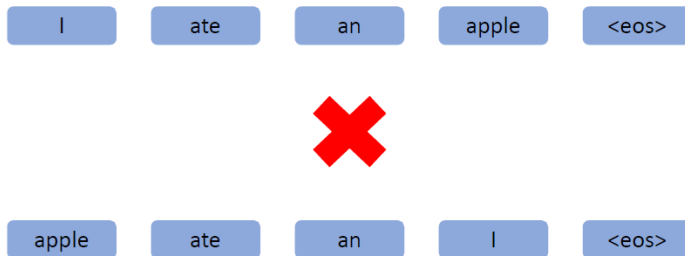




Input embeddings are dense vector representations of input tokens, allowing the model to

Position encodings are added to input embeddings to provide information about the position of tokens in the sequence, enabling the model to capture order and structure.

Position Encodings (cont.)



Requirements for Positional Encodings

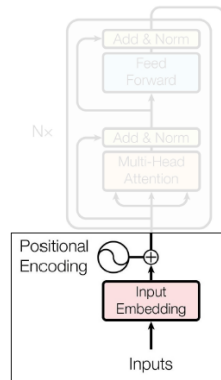
- Some representation of time? (like **seq2seq**?)
- Should be unique for each position – not cyclic
- **Bounded**

Possible Candidates

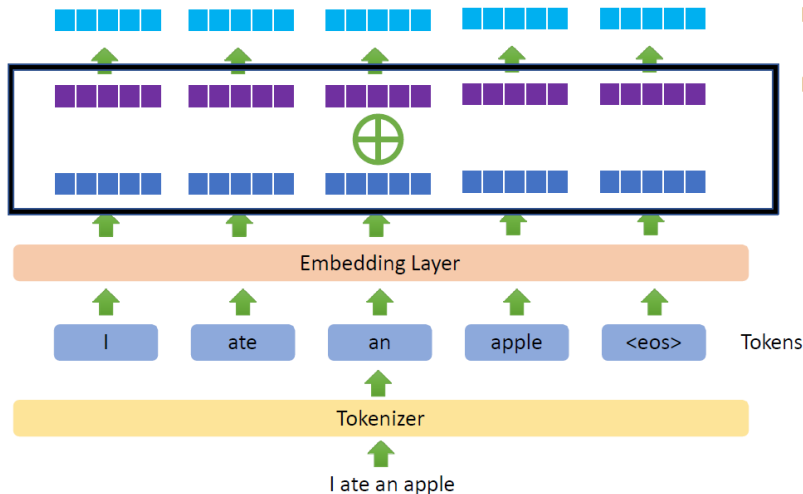
$$P(t + t') = M^{t'} \times P(t)$$

M?

1. Should be a unitary matrix
2. Magnitudes of eigen value should be 1 -> norm preserving
3. The matrix can be learnt
4. Produces unique rotated embeddings each time



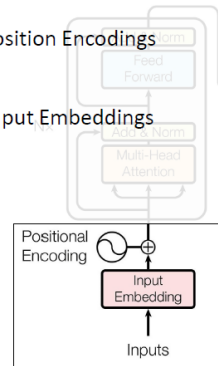
Position Encodings (cont.)

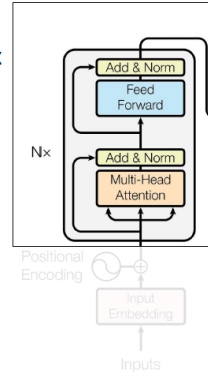
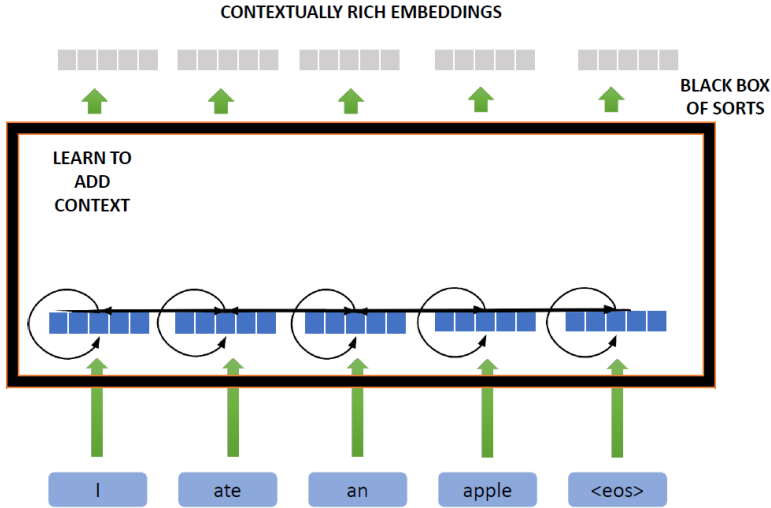


Final Input Embeddings

Position Encodings

Input Embeddings

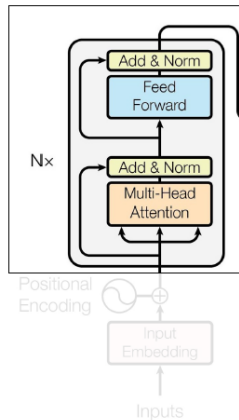




Encoder processes the input sequence, generating a set of continuous representations that

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

- Query
- Key
- Value



{Key, Value store}

{Query: "Order details of order_104"}

OR

{Query: "Order details of order_106"}

```
{ "order_100": { "items": "a1", "delivery_date": "a2", ... },  
  "order_101": { "items": "b1", "delivery_date": "b2", ... },  
  "order_102": { "items": "c1", "delivery_date": "c2", ... },  
  "order_103": { "items": "d1", "delivery_date": "d2", ... },  
  "order_104": { "items": "e1", "delivery_date": "e2", ... },  
  "order_105": { "items": "f1", "delivery_date": "f2", ... },  
  "order_106": { "items": "g1", "delivery_date": "g2", ... },  
  "order_107": { "items": "h1", "delivery_date": "h2", ... },  
  "order_108": { "items": "i1", "delivery_date": "i2", ... },  
  "order_109": { "items": "j1", "delivery_date": "j2", ... },  
  "order_110": { "items": "k1", "delivery_date": "k2", ... }
```

{Key, Value store}

{Query: "Order details of order_104"}

OR

{Query: "Order details of order_106"}

```
{
  "order_100": {
    "items": "a1",
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    ...
  },
  "order_101": {
    "items": "b1",
    "delivery_date": "b2",
    ...
  },
  "order_102": {
    "items": "c1",
    "delivery_date": "c2",
    ...
  },
  "order_103": {
    "items": "d1",
    "delivery_date": "d2",
    ...
  },
  "order_104": {
    "items": "e1",
    "delivery_date": "e2",
    ...
  },
  "order_105": {
    "items": "f1",
    "delivery_date": "f2",
    ...
  },
  "order_106": {
    "items": "g1",
    "delivery_date": "g2",
    ...
  },
  "order_107": {
    "items": "h1",
    "delivery_date": "h2",
    ...
  },
  "order_108": {
    "items": "i1",
    "delivery_date": "i2",
    ...
  },
  "order_109": {
    "items": "j1",
    "delivery_date": "j2",
    ...
  },
  "order_110": {
    "items": "k1",
    "delivery_date": "k2",
    ...
  }
}
```

Done at the same time !!

{Query: "Order details of order_104"}

OR

{Query: "Order details of order_106"}

{Key, Value store}

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  "order_101": { "items": "b1", "delivery_date": "b2", ... },  
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  "order_103": { "items": "d1", "delivery_date": "d2", ... },  
  "order_104": { "items": "e1", "delivery_date": "e2", ... },  
  "order_105": { "items": "f1", "delivery_date": "f2", ... },  
  "order_106": { "items": "g1", "delivery_date": "g2", ... },  
  "order_107": { "items": "h1", "delivery_date": "h2", ... },  
  "order_108": { "items": "i1", "delivery_date": "i2", ... },  
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```
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{ "order_101": { "items": "b1", "delivery_date": "b2", ... } },  
{ "order_102": { "items": "c1", "delivery_date": "c2", ... } },  
{ "order_103": { "items": "d1", "delivery_date": "d2", ... } },  
{ "order_104": { "items": "e1", "delivery_date": "e2", ... } },  
{ "order_105": { "items": "f1", "delivery_date": "f2", ... } },  
{ "order_106": { "items": "g1", "delivery_date": "g2", ... } },  
{ "order_107": { "items": "h1", "delivery_date": "h2", ... } },  
{ "order_108": { "items": "i1", "delivery_date": "i2", ... } },  
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{ "order_110": { "items": "k1", "delivery_date": "k2", ... } }
```

Query

1. Search for info

Key

1. Interacts directly with Queries
2. Distinguishes one object from another
3. Identify which object is the most relevant and by how much

Value

1. Actual details of the object
2. More fine grained

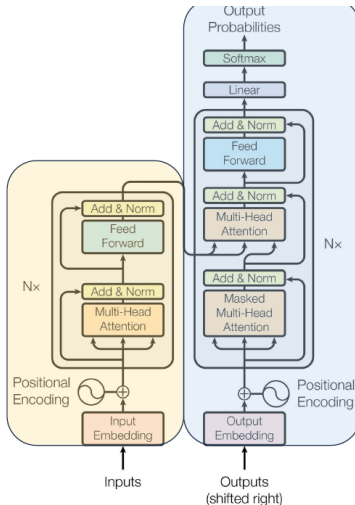
Input – input tokens
Output – hidden states

Model can see all timesteps

Does not usually output tokens, so no inherent auto-regressivity

*Can also be adapted to generate **tokens** by appending a module that maps hidden state dimensionality to vocab size*

Representation



Input – input tokens and hidden states*

Output – output tokens

Model can only see previous timesteps

Model is auto-regressive with previous timesteps' outputs

*Can also be adapted to generate **hidden states** by looking before token outputs*

Generation

NLP: **Limitations**

NLP has come a long way, but there are still some big challenges:

- ▶ **Bias:** Models can pick up and even increase unfairness from the data they learn from.
- ▶ **Needs Lots of Data:** Modern NLP needs huge amounts of labeled text, which is hard and expensive to get.
- ▶ **Ambiguity:** Language can be confusing! For example, “I saw the man with the telescope”—who has the telescope?
- ▶ **Many Languages:** It’s tough to make models that work well for every language, especially those with little data.
- ▶ **Expensive to Train:** Training big models costs a lot of money and energy, so not everyone can do it.

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